Extraction of genomic DNA from tissue culture cells

Isolating high-molecular weight genomic DNA is essential for PCR, hybridization, and other molecular analyses where degraded DNA may introduce allelic dropout or signal bias.

This protocol yields genomic DNA fragments ranging from 20–30 kbp without the need for phenol/chloroform extraction or alcohol precipitation. It is optimized for adherent or suspension mammalian cell lines. Procedures for blood, fresh or fixed tissues are not covered.

Risk assessment

– Work with human-derived material or transgenic cell lines (BSL-2)

▷ Wear gloves, safety glasses, lab coat



Reviewed: Nov 11, 2023

Procedures

›› *Preparation of cell lysates*

- ☐ 7 U/µL RNase A ⟋ R0040
- ☐ 0.6–0.8 U/µL Proteinase K
- ☐ Cell lysis buffer ⟋ R0142
- ☐ Buffer PB (Binding buffer), pH 4.8 ⟋ R0010

(1.) Prepare 10 µL proteinase K in a sterile microcentrifuge tube.

(2.) Harvest 1×10^6–5×10^6 cultured cells by trypsinization or equivalent method. Wash pellet in PBS.

> ***Hint:*** For adherent cells in a 96-well plate, wash with PBS, directly add 25 µL lysis buffer containing 2 µL proteinase K. Incubate 60 min at 55 °C in a thermocycler, then inactivate for 10 min at 95 °C. The cleared lysate can be used directly for PCR.

(3.) Resuspend pellet in 200 µL PBS and transfer to tube containing proteinase K.

(4.) *Optional:* Add 4 µL RNase A. Mix briefly. Incubate 2 min at room temperature.

(5.) Add 200 µL cell lysis buffer. Vortex to obtain a homogenous solution.

(6.) Incubate at 55 °C for 10 min to complete protein digestion.

(7.) Add 200 µL Buffer PB. Mix thoroughly.

›› *DNA purification*

- ☐ Silica membrane spin column, silica
- ☐ Buffer PE (Wash buffer) ⟋ R0011
- ☐ 5 mM Tris-Cl, pH 8.0 (⟋ R0056 for 1 M)
- ☐ Microcentrifuge tube

(1.) Apply the lysate on a silica membrane spin column. Centrifuge at 11 000 × *g* for 1 min or use a vacuum manifold.

(2.) Wash the silica membrane with 650 µL Buffer PE.

(3.) Repeat the wash step once more.

(4.) Dry membrane by centrifugation at 11 000 × *g* for 2 min.

(5.) Place the spin column into a clean microcentrifuge tube.

(6.) Apply 25–200 µL 5 mM Tris pH 8.0. Incubate 1 min. Elute by centrifugation at 11 000 × *g* for 1 min.

Storage of genomic DNA

(1.) Store at 4 °C or at room temperature for up to six months under slightly basic conditions (pH 8.0) or at −20 °C for long-term storage.

> ***Hint:*** Genomic DNA tolerates up to twenty freeze-thaw cycles without degradation.

> ***Quality assurance:*** Check DNA concentration after storage at 4 °C or room temperature to compensate for evaporation.

Analyses

○ Determine concentration and purity by measuring absorbance at 260 nm, 280 nm, and 230 nm.

Nucleic acid	A_{260} = 1.0	A_{260}/A_{280}	A_{260}/A_{230}
dsDNA	50 ng/µL	1.8–1.9	2.0–2.2

> ***Note:*** Extinction ratios vary with composition and pH. A_{260}/A_{280} may shift by 0.2–0.3 with pH.

○ Resolve 100 ng DNA on a 0.8–1.0% agarose TAE gel to verify size and integrity.

Troubleshooting

DNA purification

In Step 6:

– Low DNA recovery
  ◦ Use low-salt elution buffer between pH 7.0 and pH 8.5.

Materials

Preparation of cell lysates

- ☐ **RNase A**, 7 U/µL ◇ R0040
- ☐ **Proteinase K**, 0.6–0.8 U/µL
- ☐ **Cell lysis buffer** ◇ R0142
  Store at room temperature.
- ☐ **Buffer PB (Binding buffer)**, pH 4.8 ◇ R0010

DNA purification

- ☐ **Silica membrane spin column**, silica
- ☐ **Buffer PE (Wash buffer)** ◇ R0011
- ☐ **5 mM Tris hydrochloride (Tris-Cl)**, pH 8.0 (◇ R0056 for 1 M)
- ☐ **Microcentrifuge tube**

Recipes

Cell lysis buffer

Amount	Ingredient		Stock	Final
500 µL	Tris hydrochloride (Tris-Cl), pH 8.0	R0056	1 M	10 mM
100 µL	EDTA, pH 8.0	R0017	0.5 M	1 mM
250 µL	Triton™ X-100	R0057	10%	0.05%
	Skin irritation (H315); Serious eye damage (H318); Aquatic toxicity (long-term) (H411)			
To 50 mL	Water, reagent-grade			

Store at room temperature.

Cell lysis buffer

10 mM Tris-Cl, 1 mM EDTA, 0.05% Triton™ X-100

No GHS hazards at the indicated concentrations

☐ Hold for EHS review before disposal

Date: Sign: R0142

Change log

2017-01-01	Shany Koren-Hauer	Original protocol; Mohamed Bentires-Alj lab.
2023-11-11	Benjamin C. Buchmuller	Adaptation as SOP; supplemented from the Invitrogen "PureLink Genomic DNA Mini Kit" user manual (rev. 2007).

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